

Practice Sheet - 1

Lecture 1: Chapter 1 and 2

*Solutions are at the end of the document. SOLVE IT YOURSELF first.

1. **Explain** the three-schema-architecture with an illustrative diagram.
2. What is Data Independence? **Explain** the different types of data independence with examples.
3. Consider the following scenarios and **explain** whether they achieved logical data independence or physical data independence:
 - a. You added a new column called “age” in the employee table of the company database. The company uses an Employee management system for attendance, leaves, salary, projects etc. The “age” information will not appear to any users using the system.
 - b. Your company is upgrading their system. They previously used HDD (hard disk drive) for data storage. Now they shifted to SSD(solid state drive) for more efficient data management. The company uses a Database System for employee attendance, leaves, salary, projects etc.
4. **State** three differences between file-based approach and database approach for data management.
5. **State** two advantages of database approach for data management.
6. Consider the following scenarios and **explain** with respect to the given information and conditions whether you should use a database approach or a file-based approach for data management:
 - a. You will collect and store the birthdays and phone numbers of all your close relatives. You have around 60 close relatives. You are not concerned about security and you are the only one who will access this data.
 - b. You are hired by the government to collect and store information about all citizens in the country. The data should not be accessed by everybody. Multiple government offices will need to access different data for different purposes.
 - c. You want to store patient records, doctor schedules, and billing information for a hospital. Different departments (like emergency, radiology, and pharmacy) need access to specific parts of the data. The system must ensure data privacy and allow multiple staff members to use it and update data at the same time. (case study from slide)
 - d. You are keeping track of your daily expenses, recording the date, category, and amount spent. This data is only for your personal use, and you don’t need to apply complex queries and high security is not needed. (case study from slide)
7. **Explain** Data Abstraction.
8. **Explain** Data Model. State and Explain the different types of Data Models.

-----Go through solution only after you tried yourself-----

IMPORTANT NOTE: Paraphrasing or just reiterating what is written in the question itself, is not an appropriate explanation. So, if any question requires an explanation, it must elaborate beyond what is written in the question.

SOLUTIONS

Answer 1, 2, 4, 5, 6, 7, 8:

Use slides for explanation and own examples where examples are required.

Answer 3:

- a. Logical Data Independence. Adding an age column changes the conceptual schema, but since it does not appear to users, the external schema remains unchanged.
- b. Physical Data Independence. Changing the data storage type will change the internal schema, but the database structure remains unchanged, so the conceptual schema remains unchanged.

Answer 6:

- a. File based approach. Small amount of data that is not expected to change much. High security not required and multiple user access not required. The database approach has additional financial costs, it is not required to use a database approach since a file based approach is sufficient.
- b. Database approach. Authorized access required, large amounts of data that may change frequently, multiple user access with different views and functionality required. File based approach is not sufficient to meet the above requirements.
- c. Database Approach. Need to ensure that same info is not repeated, like patient records, so reduced redundancy by data sharing. Different dept/users will have different views of the data, which is only supported by db approach. High security required so that no unauthorized access and concurrency control is required so that data remains valid even when updated by multiple users.
- d. File-based approach. Data sharing is not needed, so no need for concurrency control or multiple different view support. High security is also not required. Thus, the additional cost of developing and maintaining a database system is not required.